Cloud & AI project

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A live flight-tracking web app: a world map of aircraft from the OpenSky Network, with ML models trained on historical flight data and an LLM layer on top.

Data

- **Zenodo (OpenSky COVID-19 flight list):** one row per flight with origin, destination, times and aircraft. Used for training.
- **OpenSky live API:** positions of all aircraft worldwide every ~90 s (position, altitude, speed, heading, vertical rate).
- **Weather (METAR airport reports):** live and historical, used as model features.

Features

- **Airspace anomaly detection:** flags map areas with unusually high or low traffic for that time of week.
- **Arrival time prediction** for flights heading to Vienna, Bratislava and Budapest.
- **Destination prediction** for live flights.
- **LLM:** a regularly updated airspace briefing, plus a chat that answers plain-English questions by querying our database.

Visuals

Dark world map with planes coloured by altitude and moving smoothly between updates, an anomaly heatmap layer, an event feed, an arrival-time board, and an AI briefing and chat panel.

Stack and CI/CD

Data collector → Postgres (TimescaleDB, PostGIS) → model training tracked in MLflow → FastAPI → React with MapLibre and deck.gl. Docker; GitHub Actions for tests, builds, deploys and scheduled retraining.

Risks

- OpenSky may block cloud server IPs, so the collector would run at home and send data to the cloud.
- Free API limits: the world map refreshes only every ~90 s.
- Anomaly detection needs 2–4 weeks of collected data first, so collection starts early.